

Exp No: 26

TRANSACTION LEVEL MODELING (TLM) USING SYSTEMC

AIM:

To model and simulate a simple Transaction-Level Model (TLM) using Initiator and Target modules in SystemC.

PROGRAM:

```
#include <systemc>
#include <tlm>
#include <tlm_utils/simple_initiator_socket.h>
#include <tlm_utils/simple_target_socket.h>

using namespace sc_core;
using namespace tlm;
using namespace std;

// Target
struct Target : sc_module {

    tlm_utils::simple_target_socket<Target> socket;

    SC_CTOR(Target) : socket("socket") {
        socket.register_b_transport(this, &Target::b_transport);
    }

    void b_transport(tlm_generic_payload& trans, sc_time& delay) {
        int* data = (int*) trans.get_data_ptr();
        cout << "Target received: " << *data << endl;
        delay += sc_time(10, SC_NS);
    }
};

// Initiator
struct Initiator : sc_module {

    tlm_utils::simple_initiator_socket<Initiator> socket;
```

```
SC_CTOR(Initiator) : socket("socket") {  
    SC_THREAD(thread_process);  
}
```

```
void thread_process() {  
    tlm_generic_payload trans;  
    sc_time delay = SC_ZERO_TIME;
```

```
    int data = 555;
```

```
    trans.set_data_ptr(reinterpret_cast<unsigned char*>(&data));  
    trans.set_data_length(sizeof(data));
```

```
    cout << "Initiator sending: " << data << endl;  
    socket->b_transport(trans, delay);
```

```
    wait(delay);
```

```
}
```

```
};
```

```
int sc_main(int argc, char* argv[]) {
```

```
Initiator initiator("initiator");
```

```
Target target("target");
```

```
initiator.socket.bind(target.socket);
```

```
sc_start();
```

```
return 0;
```

```
}
```

